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Huang, Ya; Zhou, Jingkun; Ke, Pingchuan; Guo, Xu; Yiu, Chun Ki; Yao, Kuanming; Cai, Shaoyu; Li, Dengfeng; Zhou, Yu; Li, Jian; Wong, Tsz Hung; Liu, Yiming; Li, Lei; Gao, Yuyu; Huang, Xingcan; Li, Hu; Li, Jiyu; Zhang, Binbin; Chen, Zhenlin; Zheng, Huanxi; Yang, Xingyu; Gao, Haichen; Zhao, Zichen; Guo, Xu; Song, Enming; Wu, Hui; Wang, Zuankai; Xie, Zhaoqian; Zhu, Kening; Yu, Xinge

Published: 01/12/2023

Document Version:

Post-print, also known as Accepted Author Manuscript, Peer-reviewed or Author Final version

Publication record in CityUHK Scholars:

[Go to record](#)

Published version (DOI):

[10.1038/s41928-023-01074-z](https://doi.org/10.1038/s41928-023-01074-z)

Publication details:

Huang, Y., Zhou, J., Ke, P., Guo, X., Yiu, C. K., Yao, K., Cai, S., Li, D., Zhou, Y., Li, J., Wong, T. H., Liu, Y., Li, L., Gao, Y., Huang, X., Li, H., Li, J., Zhang, B., Chen, Z., ... Yu, X. (2023). A skin-integrated multimodal haptic interface for immersive tactile feedback. *Nature Electronics*, 6(12), 1020-1031. <https://doi.org/10.1038/s41928-023-01074-z>

Citing this paper

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<https://doi.org/10.1038/s41928-023-01074-z>.

A skin-integrated multimodal haptic interface for immersive tactile feedback

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Abstract:

Haptic feedback can be used to improve immersion in virtual and augmented reality systems. However, wearable haptic devices typically provide limited feedback, such as simple vibration, and are restricted to a narrow bandwidth of tunable frequency or to single tactile perceptions. Herein, we report a skin-integrated wireless haptic interface that is based on actuator arrays and offers multimodal and complex feedback mechanisms. Different feedback modes — mechanical, electrotactile and thermal — in the interface are used to selectively activate different cutaneous receptors, providing users with diverse haptic sensations. The selective stimulating mechanism is inspired by the neural tactile sensing system and can reproduce tactile information of fine roughness, macro roughness, slipperiness, force and temperature.

Tactile sensations involve the acquisition of complex and coupled data via mechanoreceptors in the skin, and are difficult to accurately reproduce using haptic feedback interface.¹⁻³ Various hardware approaches for haptic feedback have been explored, including grounded devices⁴, active surfaces⁵, handles⁶, and wearable devices³. Wearable devices offer the highest mobility and potential for interaction with the real world (Supplementary Fig.1). In addition, devices that form a conformal contact with the skin can minimize the impact on physical interaction and offer an efficient information transmission path between users and devices.^{7,8} However, skin-integrated haptic interfaces are still in the early stages of development.⁹

Tactile perception can be classified into two major subsystems: one for temperature, detected by free nerve endings and Krause end bulbs in the skin; and one for mechanical stimuli, detected by Meissner's corpuscles, Merkel's disks, Pacinian corpuscle and Ruffini endings (Fig. 1a).¹⁰⁻¹² Tactile sensation is associated with multidimensional factors, including fine roughness, macro roughness, hardness, slipperiness and temperature.¹³ Therefore, realistic haptic reproduction requires a feedback interface that includes as many of these dimensions as possible.¹⁴⁻¹⁶

Temperature sensation depends on the heat flux that occurs when touching different objects, which can be related to the temperature and thermal properties of different surfaces.¹⁷ Active cooling and heating effects place high demands on the heat conductivity and heat dissipation capability of wearable devices.¹⁸ Mechanical tactile feeling, which senses surface textures and forces, is decoded from spatiotemporally distributed pressure and vibration. For a haptic interface, the frequency should be tunable within the sensitive range of human tactile sensations, which is 0-250 Hz.¹⁰ Research on frequency-leveraged actuators that can generate high output displacement even in small volumes has focused on resonance based on piezoelectric actuators¹⁹, linear resonant

actuators²⁰ and eccentric rotating mass motors²¹. However, the amplitude of the displacement created by these actuators is coupled with the frequency, and greatly decreases outside of their resonant frequency.⁷

Pressure sensation usually requires large forces due to the strong damping effect of the skin, which is challenging to create with miniaturized actuators used for vibration.²² Pneumatic and hydraulic systems are suitable for producing static pressure, but are not high-frequency vibration due to their low response speed.²³ Since nerve activity codes all sensations, it should in theory be possible to create versatile tactile modalities by mimicking nerve signals.^{24,25} Electrotactile displays have been proposed as an alternative to conventional mechanical tactile displays without considering mechanical resonance.^{26,27} However, electrotactile modes are not precise enough to stimulate receptors located deep in the skin, such as Pacinian corpuscles that sense high-frequency vibration.²⁸ In addition, electrotactile stimulation triggers a “harder” or “sharper” sensation, since all the receptors that receive a sufficient current fire together.²⁹

In this article, we report a skin-integrated multimodal haptic feedback interface that can stimulate different sensory receptors in the skin based on the three actuation modes of temperature, mechanical deformation and electrotactile (Fig. 1a). Thermoelectric stimulation is used to control temperature changes on the skin, which reproduces the thermal feelings between the skin and objects. The sandwich structure of the devices achieves high heat dissipation to ensure good thermal feedback for activating free nerve endings or Krause end bulbs. Mechanical sensations are achieved by creating skin deformations using mechanical actuators, targeting the deep touch sensed by Pacinian corpuscles and Ruffini endings,⁷ and electrotactile stimulation is used to create nervous activity of the shallower Merkel's disks and Meissner's corpuscles (Fig. 1b).^{14,30,31} The

interaction between electrotactile and mechanical vibrators can enhance their integrated performance compared to a single mode, improving the fidelity of virtual mechanical tactile sensations over a frequency range of 0-250 Hz. We test the capability of the developed multimode haptic feedback interface when located on a person's wrist, arm, and palm by comprehensive user studies, and can create virtual tactile sensations of fine roughness, macro roughness, slipperiness and temperature.

Mechanism of multimodal tactile sensation stimulation

The haptic feedback unit operates with three feedback modes: thermal feedback based on two pairs of thermoelectric pellets, mechanical feedback based on a miniaturized mechanical actuator and electrotactile feedback based on thin and soft Au electrodes (Supplementary Fig. 2). The difference in thermal conductivity coefficient between skin and the targeted object results in the perceptions of warm or cold feeling for different materials. Extended Fig. 1 shows the temperature changes simulated when human fingers touch different materials, where the temperature change is significantly influenced by the thermal conductivity coefficient of the targeting materials. For example, touching 2 mm thick brass (thermal conductivity coefficient, $k = 116 \text{ W/mK}$) causes a temperature change over 7°C in 5 seconds of touching and over 8°C after 30 seconds of touching, while the temperature change is less than 3°C for touching PDMS (2 mm thick, $k = 0.15 \text{ W/mK}$) and less than 2°C for touching cotton ($k = 0.03 \text{ W/mK}$) even after 30s (Supplementary Fig.3). Additionally, the temperature change can vary according to changes in the geometric size of materials or skin (Supplementary Fig.4 and Fig.5). In our device, the thermoelectric pellets modify the interface temperature between the top layer of the device and

skin according to the virtually touched materials in the range of 20-45 °C (Fig. 1c). The device produces active cooling and heating sensation based on the Seebeck effect.¹⁷

Vibratory cues, in conjunction with spatial cues, can be utilized to perceive different tactile feelings of the surface of touched objects. To achieve mechanical tactile representation, we selected electromechanical actuators based on Lorentz force to create physical skin deformation and provide a sensation of high-frequency vibration (Fig. 1a). In this design, a polyimide (PI) sheet with the cantilever structure holds a magnet for vibration to stimulate the mechanoreceptors under the skin (Fig. 1b, Supplementary Fig.6).^{32,33} However, the vibration travel amplitude of mechanical actuators is strongly affected by the working frequency. In Fig. 1d, the maximum travel amplitude of the actuator is above 0.8 mm at the resonant frequency around 200 Hz, but the travel amplitude reduces to around 0.2 mm at 100 Hz when activated by the same voltage. The maximum force applied by the vibration is up to 50 mN at around 200 Hz, but the force reduced to around 20 mN at 100 Hz.³⁴

Electrical pulse depolarizes nerve fibers to create tactile sensations. The period of the nerve's activity synchronizes with the period of vibration sensation (Supplementary Fig.7).³⁵ To study the frequency discrimination ability of electrotactile and mechanical vibration, we conducted a user study in which participants were asked to compare a standard frequency with other frequencies lower or higher than the standard frequency. We increased or decreased the frequency from standard frequency for comparison until participants could distinguish them with standard frequency, and the threshold frequency difference at different standard frequency is shown in Fig. 1e. We observed that the frequency difference of electrotactile is small at low frequency (within 20%), especially below 80 Hz. However, the discrimination threshold of electrotactile began to

increase for frequency beyond 80 Hz, and participants were even difficult to discriminate 120 Hz and 200 Hz in electrotactile. On the other hand, the lower and upper discrimination thresholds of mechanical vibration below the standard frequency of 140Hz are still within 20% and 30%, respectively.

Meissner's and Pacinian corpuscles are two types of mechanical receptors which respond to mechanical vibrations. Meissner's corpuscles are located in the shallow dermis near epidermis, while Pacinian corpuscles reside in the hypodermis or deep part of dermis.^{10,11} They can be activated at their typical sensitive frequency, where Meissner's corpuscles are stimulated by the vibration below 100 Hz (most sensitive between 20~40 Hz) and Pacinian corpuscles are activated higher than 80 Hz (most sensitive between 200~250 Hz).²⁹ However, electrotactile stimulation is considerably more effective in stimulating the superficial nerves, as evidenced by the electrical current and field distributions simulated beneath the skin (Extended Fig. 2 and Supplementary Fig.8). As a result, Meissner's corpuscles are inevitably co-activated in high-frequency electrotactile due to the stronger electrical pulse depolarization in the shallow region of the dermis. The brain misinterprets the activation of Meissner's corpuscles with Pacinian corpuscles at high-frequency electrotactile, leading to difficulties in frequency discrimination. To overcome this limitation, we propose a hybrid method of applying mechanical and electrotactile simultaneously, where mechanical actuators selectively stimulate Pacinian corpuscles while electrotactile targets Meissner's corpuscles. (Fig. 1c).

Users can feel a force applied to the skin in each pulse of electrotactile and it is easy for them to discriminate each pulse in the frequency range lower than 10 Hz. An electric pulse with high intensity can activate a vigorous impulsive force on the skin, which can be used to mimic some punching behaviors in virtual situations (Fig. 1f). With the increase of frequency, applied force

with short intervals is easy to feel like vibration on the skin. Furthermore, when users feel a continuous force (either in constant or varied format) applied on their skin, dynamic pressure is created. Typically, the continuous pressure sensation is activated by electrotactile when skin receives continuous force and cannot discriminate each pulse (Fig. 1f).³⁶ The dynamic pressure activated by electrotactile is properly composed of slight sensations of vibration, like skin sliding over a hump (Fig. 1c).³⁷

In our multimode haptic feedback interface, electrotactile mainly stimulates Merkel's disks and Meissner's corpuscles to create the tactile feelings of pressure, vibration and impulsive force (Fig. 1c).^{14,34} A strong electric pulse can activate an impulsive force on the skin, which can be used to reproduce some dynamic interactions between skin and objects. The impulsive force created by electrotactile is much higher than that provided by mechanical actuators. Electrotactile and mechanical deformation can work together to provide a wide frequency range of vibration related to tactile sensation on different surfaces. Integrated with thermoelectric actuators, the whole devices can effectively provide multiple tactile sensations, including punching, vibration, pressure/hump, warmth and cold.

Design of multimodal haptic interface

The integration of thermal feedback, mechanical vibration and electrotactile in skin-integrated interfaces can provide diverse tactile sensations to users in VR/AR while preserving their ability to interact with the real world (Fig. 2a). The right part of Fig. 1a shows the schematic illustration of the versatile haptic interface, where the three types of actuators in array formats, electrical interconnects, thermal insulation fibers and the soft thermal conductive substrate are integrated. In the haptic interface, a thin layer of laser-cut Cu/PI (18 μm /12.5 μm), supported by

a soft silicone thermal conductive substrate (mixed with carbon compounds and metal oxide compounds, 300-1000 μm thick, Supplementary Fig. 9), serves as electrical interconnects to afford compact route for closely spaced mechanical actuators and thermoelectric pellets (Supplementary Fig. 2). After mechanical actuators and thermoelectric pellets are welded, thermal insulation ceramic microfibers fill in the gaps among all the mechanical actuators and thermoelectric pellets to inhibit the heat flow back (Supplementary Fig. 10). Another layer of thermal conductive thin film with an opening on each mechanical actuator encapsulates the device and serves as the supporting layer for electrotactile electrodes. The mechanical property of encapsulation layer has no obvious effect on the resonance frequency of mechanical actuators (Supplementary Fig. 11). Photo-defined Cr/Au/PI thin film (5 nm/200 nm/2 μm) patterns act as soft electrotactile circuits. Another layer of PI (2 μm thick) with selective dry etching allows for encapsulating the interconnects and leaving the electrode open to contact with the skin. The multimodal haptic feedback interface can be designed and customized into various shapes with different actuator arrays. Figs. 2b-d show three types of devices: a strip with a 2 $\times$ 4 feedback units array, a patch with a 4 $\times$ 4 feedback units array and a palm format with distributed actuators over the hand, respectively. Their detailed dimension and weight of different devices are summarized in Supplementary Table 2. These devices can be directly interfaced with various body locations such as wrist, arm, palm and other positions (Supplementary Fig. 12). The electric circuits on different layers remain stable even after being bent for 500 cycles. (Supplementary Fig. 13)

The multimodal haptic feedback patch can be powered by coin-size batteries (Supplementary Fig. 14) and wirelessly operated by thin and miniaturized wristband-type flexible circuits (Fig. 2e). Extended Fig. 3 shows the block diagram of the control circuit for the multimodal haptic feedback

interface. The micro-controller unit (MCU) communicates with personal computers (PCs) through a Bluetooth Low Energy (BLE) module to receive the commands from the PC and generate corresponding electrical signals to drive the three feedback modules. The voltage conversion module consists of a low dropout regulator (LDO), a boost converter and a buck converter. The LDO steps down the input voltage to 3.3 V to power the MCU, while the boost converter module steps up the input voltage into high output voltage to drive the electrotactile module. The buck converter is used to convert the input voltage to a low level of 2.5 V to enhance the current output capability, which can drive the actuator and thermoelectric modules. The operation of the thermal feedback is associated with controlling the input power for the thermoelectric pellets with either positive or negative voltage to realize heating or cooling.

It is challenging for skin-integrated devices to have a good cooling effect due to their limited heat dissipation capability, resulting in an obvious temperature increase in cooling process caused by residual heat. To improve the thermal dissipation performance, a sandwich structure is designed (Supplementary Fig. 15). Two sides of the device are encapsulated by soft thermal conductive (TC) materials with a relatively high thermal conductivity coefficient ($k \sim 1.93 \text{ W/mK}$). The inner part between the two TC layers is filled with a thermal insulation material (ceramic microfibers, $k \sim 0.035 \text{ W/mK}$) to reduce heat conduction between the two TC layers. The sandwich design can maintain a large temperature difference between the two TC layers, which is confirmed by simulation results and detected infrared photographs (Supplementary Fig. 16). Fig. 2f shows time-dependent temperature profiles of both cooling and heating modes on a 4×4 patch under various power inputs. Both simulation and experiment results indicate that the device can produce desirable temperature changes for cooling and heating rapidly (Extended Fig. 4 and Supplementary

Fig. 17). To investigate the active cooling effect and heating effect, the temperature change progress is simulated for the device mounted on skin (Extended Fig. 5), where the heat dissipation progress occurs on the TC layer contacting with air (outer TC layer). Fig. 2g shows the influence of the thickness and heat dissipation structure (HDS, Supplementary Fig. 18) of the outer TC layer in the cooling progress on the interface between the device and skin. Increasing the thickness of the outer TC layer and adding the HDS can improve the cooling effect and inhibit temperature increase caused by residual heat. Moreover, higher air convective heat transfer can enhance the cooling effect due to the better heat dissipation (Supplementary Fig. 19). On the other hand, increasing the thickness of the outer TC layer enhances the heating capability of the interface adhered to the skin (Supplementary Fig. 20). For a 1 mm thick outer layer, the temperature decreases over 7 °C, showing a negligible increase within the following 10s in the cooling progress (Extended Fig. 5). An embedded design of the outer TC layer can effectively reduce the thickness of the device without reducing the heat dissipation effect (Supplementary Fig. 21). Finally, the device can achieve a maximum ΔT over 7 °C for the cooling mode and 12 °C for the heating mode with the input current around 1 A within 5 s (Fig. 2h).

Tactile sensation reproduced by three feedback modes

A group of volunteers participated in a test in which they reported their feelings of touching different materials by feeling the different thermal properties and comparing them with real materials (Fig. 2i). In the cooling mode with a power density of around 100~130 mW, the participants could easily feel chilly ($\Delta T > 4.5$ °C), which convinced them of touching materials with a thermal conductivity coefficient > 1 W/mK, such as quartz glass ($k = 1.4$ W/mK), stainless steel (SS, $k = 25$ W/mK) and brass ($k = 116$ W/mK). Reproducing the tactile sensation of thermal

insulation materials is more complicated than that of thermal conductive materials. Computational results reveal that materials with thermal conductivity coefficient < 0.1 W/mK may also result in a slight temperature decrease (Extended Fig. 1). Considering the microstructures and textures of the materials, the physical touch feeling of these materials such as cardboard ($k = 0.07$ W/mK), cotton cloth ($k = 0.06$ W/mK), and plastic foam ($k = 0.03$ W/mK) is often warm.³⁸ Due to the high thermal conductivity coefficient of our encapsulation layers, the devices should heat up slightly to mimic the heat flux change when skin touch materials with low thermal conductivity coefficient (Supplementary Fig. 22). Therefore, the warming mode relies on the combination of theoretical data and extensive user studies (Fig. 2i).

We simulated the behavior of a finger sliding across surface with varying levels of fine roughness (Fig. 2j and Extended Fig. 6), where the roughness perceptions resulting from vibratory stimuli are sensitive to dynamic stimulation caused by finger movement. The change in stress frequency on the finger is strongly affected by the sliding speeds and the surface roughness (Fig. 2k). Surface with lower roughness represents smaller spacing among particles, which can lead to higher frequency responses during sliding owing to the smaller intervals as the sliding speed increases. On the same roughness surface, a sharper morphology can cause a larger stress (Fig. 2l). We tune the frequency and amplitude of electrotactile to activate different low-frequency vibrations to reproduce the sensation of fine roughness. Fig. 2m presents the results in which participants compared the stimulated roughness on the electrode with the real tactile sensation of touching sandpapers. The results reveal that higher frequency and lower amplitude of electrotactile can reproduce the tactile sensation of touching sandpapers with smaller roughness.

When the intensity of electrotactile is low, the sensation of electrotactile is similar to that of mechanical stimulation. Sometimes, participants cannot even tell which was electrotactile and which was mechanical stimulation. However, when the intensity becomes high, they can distinguish the sensation of electricity clearly. Mechanical stimulation offers a softer feeling, while electrotactile brings a stronger or sharper feeling. This may be because the electricity stimulates localized nerves synchronously, while mechanical deformation leads the force to be distributed to a large area, stimulating surrounding mechanoreceptors with different intensities.³⁹ Thus, mechanical vibration is more suitable for stimulating the tactile sensation when a finger moves across a soft surface, and electrotactile can be added to increase the “harder” roughness in some virtual textures. In addition, vibration activated by electrotactile and actuators can be programming controlled based on different texture patterns. Fig. 2n shows four texture patterns of soft materials reproduced by the mechanical actuators array (Supplementary Fig. 23), which is easy for participants to distinguish the shape (Fig. 2o).

Tactile feeling obtained from the macro morphology of different surface, such as humps, are closely related to pressure sensations. As we discussed earlier, undistinguished pulse can produce continuous pressure sensation on the skin to mimic humps. We conducted a user study to rate the pressure sensation activated by three kinds of electrotactile, including continuous electrotactile, periodic electrotactile with periodic cycles of 2s and 1s (Fig. 2p). The average rating levels of pressure sensation in these three modes at different frequency follow a similar pattern. The rating levels increased sharply with the increase of frequency lower than 80 Hz but slightly reduced due to the increase of tickle or tingling feeling caused by electrotactile over 100 Hz. We collected the frequency at which participants rated the highest level of pressure, mostly ranging from 40-100 Hz

as shown in Fig. 2q. We point out that the surface morphology can be divided into fine roughness and macro roughness. In the dynamic touching behaviors where the skin and object are in relative motion, fine roughness can be represented by the tunable vibration produced by mechanical actuators and electrotactile, and macro roughness can be represented by the period activation of electrotactile (Supplementary Fig. 24). Therefore, changing the portfolio of the activated actuators allows to realize both macro roughness and fine texture sensations with a static contact between the device and skin.

To further compare the resolution and density of our skin-integrated haptic feedback interface with other published works based on different haptic feedback mechanisms, we summarized the geometric size, density and functions of different systems in Supplementary Table 4.

Texture recognition with multimodal haptic interface

When we touch real objects, our skin gathers diverse information about the interactions, which is crucial for decoding the tactile sensations. The richness of these interaction cues is often missing from many single-mode haptic feedback interfaces, resulting in a less satisfying and immersive experience. Feeling multi-dimensional tactile sensation from haptic feedback interfaces can create the perceptual illusion that one is touching a real surface. To evaluate the virtual tactile perception production ability of our multimode haptic interfaces, we conducted a series of user studies to evaluate the virtual tactile perception produced by our devices. Fig. 3a shows nine typical materials including fabric, sandpapers, plastics, and SS used in our user study, which have diverse fine textures, morphology and patterns. Wool fabric is a soft material often used for clothing, which is always felt warm when touched. In contrast, SS is a rigid metal that causes an obvious temperature decrease when touched.

Virtual textures rendered with the data-driven method are more realistic than those generated with periodic signals. To prove the reappearance ability of our haptic feedback interface and convert the texture signal of different surfaces into digital signals that can be processed and transmitted by a computer, we recorded pressure signals and acceleration signals from real touching behaviors to create texture models (Supplementary Fig. 25).⁵ In practical situations, feelings of fine texture information can be achieved by moving fingers over the surface of an object to get a vibration sensation. In our haptic interface, the feedback of fine textures can be reproduced by adjusting the pulse width and the frequency of electrotactile and mechanical vibration to simulate the intensity of the force and discriminate the vibration. The acceleration signal is used to analyze the small vibrations generated by the micro morphology of the materials during finger sliding, which is related to the fine roughness and slipperiness of the surface (Supplementary Fig. 26). The mechanical actuator exhibits high resolution at high frequencies and can be used to reproduce the high-frequency vibration signals collected by accelerometers. Real vibration signals can be encoded by pulse width modulation (PWM) signals to drive mechanical actuators to mimic the real vibration (Supplementary Fig. 27). Low-frequency vibration signals can be reproduced by tunable electrotactile. As the finger passes over different surfaces, the macro morphology of textures can be reflected in the periodic change of pressure, like T3, T6, T7, T8 and T9 (Supplementary Fig. 28). In order to produce a sensation similar to the pressure change on the skin, we determine the period cycle of electrotactile based on the full width at half maximum (FWHM) of the pressure peak. The intensity of each cycle of electrotactile is controlled by the pressure value. Then the macro morphology related to pressure sensation is generated by periodic electrotactile for mimicking the behavior of skin sliding across these humps (Supplementary Fig. 29).

We conducted a within-subject study to evaluate the performance of tactile simulation using our device. We designed a user-perception experiment with two separated parts, one for subjective rating and another for tactile matching.⁴⁰ In the first subjective rating experiments, participants placed their palms firmly on the multimodal haptic feedback interface to perceive the virtual texture during the study. The haptic feedback stimuli of nine textures were individually provided to the participants at one time in randomized order. The stimuli conditions (i.e., real or virtual) were in a counter-balanced order. 15 participants completed the adjective ratings for 18 textures (9 real, 9 virtual) in five dimensions, fine roughness (smooth-rough), macro roughness (flat-lumpy), slipperiness (slippery-sticky), hardness (soft-hard), and temperature (cold-warm) (on a scale from 1 to 7), which were chosen based on the psychophysical dimensions applicable to the perception of textures.⁴¹ The participants rated all textures along each scale before proceeding to the next scale. The average rating levels in these five dimensions of each texture were shown in the radar chart of Fig. 3a, where the levels of virtual texture showed a similar shape to that of real texture in most textures. The original adjective rating data in five dimensions are shown in the box plots of Figs. 3b-3f, where real textures and virtual textures are listed in pairs. A two-factor repeated-measures analysis of variance (ANOVA) was performed separately on the five sets of adjective ratings using texture and virtual/real as factors.⁴⁰ When comparing across textures, the differences in rating of smooth-rough ($F(8, 352) = 102.204, p < 0.001$), flat-lumpy ($F(8, 352) = 86.577, p < 0.001$), slippery-sticky ($F(8, 352) = 38.543, p < 0.001$), soft-hard ($F(8, 352) = 240.158, p < 0.001$), and cold-warm ($F(8, 352) = 428.854, p < 0.001$) were found to be statistically significant. When comparing virtual and real texture, the differences in smooth-rough ($F(8, 352) = 0.402, p = 0.529$), flat-lumpy ($F(8, 352) = 2.150, p = 0.150$), slippery-sticky ($F(8, 352) = 2.828, p = 0.100$) and cold-warm ($F(8, 352) = 0.194, p = 0.662$) were not found to be statistically

significant, proving that our haptic feedback provide good virtual tactile sensations in these four dimensions. However, the difference in soft-hard was found to be statistically significant ($F(8, 352) = 25.487, p < 0.001$), and the rated hardness of the virtual textures did not correlate well with the real textures. Previous studies have reported that users found it difficult to discern the hardness of materials when they touched a flat surface, but could clearly feel both unevenness and hardness of the same material when they perceived a change in shape or morphology on the surface. The changes in friction sensation are related to the sliding acceleration of fingers on the surface, and sometimes participants were unable to clearly distinguish the difference between friction and fine roughness when they moved their fingers on a surface.¹⁴ Stimulation of high-frequency components corresponding to the acceleration of hand movements could produce a natural sliding feeling.³⁹

In the tactile matching section, virtual textures activated by three conditions, only thermal feedback, only mechanical feedback (mechanical deformation and electrotactile), or both thermal feedback and mechanical feedback (mechanical deformation and electrotactile), were given to participants in a randomized order. Then participants were asked to choose the most similar real materials based on their haptic sensations obtained from the feedback device. Fig. 3g shows the test results of distinguishing these materials only by temperature cues, where T1-3, T4-6, and T7-9 are three groups that couldn't be distinguished. Fig. 3h shows the results of distinguishing these materials only by mechanical information produced by mechanical actuators and electrotactile. Compared to the single mechanical haptic feedback, the multimodal haptic interface allows users to achieve higher accuracy in recognizing different materials (Fig. 3i). The improvement of 18%,

31%, 31%, 35%, 11%, 45%, 24%, 36% and 20% for T1 to T9 textures, respectively, demonstrate the advantages of producing rich tactile information in multimodal haptic feedback interfaces.

Integration of haptic interface with VR applications

Fig. 4a demonstrates the use of the multimodal haptic interface for experiencing interactions with the virtual world in immersive VR applications (Supplementary Video 1). The user wears a VR headset and three soft haptic feedback patches on her wrist, arm and hand. The commercially available VR head-mounted device (HMD) provides visual/auditory information and hand motion captures. The haptic feedback patches, synchronized with the HMD, generate tactile sensations from the video contents. The video and auditory contents are created by unity 3D Game. Due to the skin-integrated design of our haptic feedback interface, it is easy for the engine synchronized with the motion controller of the HMD to trace the user's hand movements, detects collisions between the virtual hand and the virtual object, and send commands to the haptic interfaces for corresponding tactile feedback. We demonstrate applications in both indoor and outdoor scenes. For the indoor scenes, several materials, including a wet tissue with 75% alcohol, a notebook with smooth and rough surface areas, a wooden tabletop, and a cup with hot water, are presented for the user to interact with (shown in Figs. 4b, 4c and Extended Fig. 7). For the outdoor scenes, several different types of weather, including breeze, strong wind, heavy rain, light rain, as well as the sensation of a falling insect, are demonstrated (Fig. 4d, 4e and Extended Fig. 8). In addition to tactile sensation on texture, impulsive force is an effect haptic feedback in virtual reality for mimicking collisions with virtual objects. In outdoor scenes, impulse feedback is effective in reproducing the tactile sensation when raindrops fall on the skin. During the immersive experience,

users can feel dynamic changes of temperature, force, texture and vibration, which provide rich tactile sensations of the virtual world.

We further conducted a user study with four scenes for participants that associate disinfection with 75% alcohol tissue, pouring hot water into a glass bottle, strong wind and heavy rain, to rate the similarity of their haptic feedback perception with real tactile feelings and compare the difference between visual-auditory feedback and real sensation. (Extended Fig. 9) The integration of visual-auditory and haptic feedback achieved higher similarity levels than single-mode feedback. These rich interactions and multisensory feedback make it possible to bring the virtual world closer to reality, allowing users to interact with everything in their daily life and get a more immersive experience.

Conclusion

We have reported a thin and soft multimodal haptic feedback interface that can be directly integrated with the skin. The device incorporates thermal, mechanical and electrical feedback modes that target different receptors in the skin to achieve versatile haptic sensations for force, temperature and texture sensing. Evaluations in user studies of the tactile sensations created by our interface show high fidelity in the dimensions of fine roughness, macro roughness, slipperiness and temperature compared to tactile sensations produced from real objects. Hardness sensation could be further improved by increasing the density and spatial distribution of pressure feedback. The development of a skin-integrated multimodal haptic feedback interface could enable users to interact with the virtual world in a more immersive experience, with potential applications in online working, entertainment, surgical robotics, industrial training.

Methods:

Fabrication of the mechanical actuators. The diameter of the mechanical actuator is 5 mm. The ring of the actuator was firstly fabricated by 3D printing technology with UV-curable resin (inner diameter of 4 mm, outer diameter of 5 mm, thickness of 1 mm). Polyimide (PI) supporting layer (40 μm thick) was designed and laser cut to semicircular disk. A permanent magnet with a diameter of 2mm and a thickness of 0.5 mm was glued on the cantilever-like platform of the semicircular PI disk and then attached to the resin ring via glue. A coil (inner diameter of 2 mm, outer diameter of 5 mm, thickness of 0.25 mm) with 50 turns of Cu wires with a diameter of 50 μm glued on the other side of the UV-curable ring, and the cavity between the ring and the magnet allowed the magnet to vibrate freely. After assembling all parts, the overall dimension of the actuator was 5 mm in diameter and 1.5 mm in thickness, respectively.

Fabrication process of the electrotactile electrodes. The fabrication process for the flexible electrodes started on a glass substrate, which was used without any further treatments. Firstly, poly (methyl methacrylate) (PMMA) solution was spin-coated on the glass to serve as a sacrificial layer (annealed at 200 $^{\circ}\text{C}$ for 20 min), then a thin layer of PI (2 μm , annealed at 250 $^{\circ}\text{C}$ for 30 min) was spin-coated to support a thin film metallic electrode layer (e-beam sputtered Au/Cr with a thickness of 150 nm/10 nm). Then, photolithography (URE-2000B) and etching processes were applied to form the desired metallic electrodes for the electric circuits. Subsequently, another PI layer (2 μm , annealed at 250 $^{\circ}\text{C}$ for 30 min) was spin-coated on the metallic pattern, followed with photolithograph and selective Reactive Ion Etching (RIE) etching process to remove other parts and only coated the connecting lines of electric circuits are retained. Finally, the thin layer of Au/Cr/PI electrodes was transferred onto the surface of thermal conductive membrane (~ 300 μm thick).

Assembly of multimodal haptic feedback patch. Commercial thermal conductive materials (PINSAL, GK-920, silicon compounds 35%, carbon compounds 35%, metal oxide compounds 30%) were drop cast on the glass and annealed at 100 $^{\circ}\text{C}$ for 60 min to obtain the smooth thermal conductive membrane (Supplementary Fig. 9). Thermal conductive membrane with designed morphology can be fabricated by PVA modes. The fabrication process of electric circuits started on a thin Polydimethylsiloxane (PDMS) layer where a Cu foil (18 μm) coated PI (12.5 μm) was attached on it. Laser cutting was conducted to obtain the desired metallic electrode patterns, which can be transferred onto a thermal conductive layer with designed morphology, followed by soldering thermoelectric pellets and mechanical actuators on the electrodes (Supplementary Fig. 2). The unique mechanism of the thermoelectric devices required soldering n-type and p-type pellets (bismuth telluride, 1.3 mm $\times$ 1.3 mm $\times$ 1.65 mm) alternatively. Each mechanical actuator was close to a pair of thermoelectric pellets in one unit. Subsequently, the surrounding of these pellets and actuators was encompassed with $\text{SiO}_2\text{-Al}_2\text{O}_3$ thermal insulation microfibers (Supplementary Fig. 10), which was prepared by blow spinning following the method of the work which has been published before.⁴² The photo-defined Au/Cr/PI electrodes layer were transferred onto another layer of thermal conductive membrane with a thin layer of uncured material. After cured, an array of holes was created aligned to the position of actuators. Finally, the thin layer of thermal conductive membrane (~ 300 μm thick) with Au/Cr/PI electrodes would be coated on top of the thermoelectric pellets and mechanical actuators.

Control circuits. There are two kinds of control circuits in our system, one is wireless control circuit which can communicate through the Bluetooth Low Energy (BLE) and powered by a

battery; the other is the USB connected circuits which communicate through Universal Asynchronous Receiver/Transmitter (UART) (Supplementary Fig. 30). The circuit contains main parts as follow: microcontroller (MCU) (Supplementary Fig. 31), communication module (UART or BLE) (Supplementary Fig. 31), voltage conversion module, electrical stimulation module, thermoelectric module and mechanical actuator module, respectively. The computer transmits the control signal to the MCU (MSP430F5237 or CC2640R2F, TI) through the communication module (2450AT18A100E or CH340E, Johanson Technology or WCH). After receiving the command, the MCU controls the electrotactile module, the mechanical actuator module and the thermoelectric module to perform the corresponding actions according to the received commands. The voltage conversion module consists of low dropout regulators (LDO), boost converters (Supplementary Fig. 31) and buck converters (Supplementary Fig. 31). LDO (TPS76933, TI) helps to step down the input voltage to 3.3V to power the MCU and communication module, while Boost converter module (LT8364 and LT8331, Analog Devices) functions as stepping up the input voltage to 60V or higher voltage to drive the electrotactile module. The power supplies are USB or batteries, and they can only supply the current lower than 1 A. The buck converter (LM3743, TI) is used to step down the input voltage to 2.5 V in order to increase the output current capability to drive mechanical actuators and thermoelectric module, thus the maximum output currents can reach 2 A. The electrotactile module contains multiple 8-channel multiplexers (MAX14752, Maxim Integrated) and multiple electrodes in contact with the skin. The MCU controls the activation of electrotactile output channels through the control ports of the multiplexer, thus generating electrotactile signals of different frequencies and duty cycles. The mechanical actuator module uses metal–oxide–semiconductor field-effect transistor (MOSFETs) as electronic switches, each of MOSFETs (CSD13380F3, TI) controls a corresponding actuator, and the MCU controls the actuator to vibrate at different frequencies and intensities by controlling the on/off of the electronic switches. The thermoelectric module consists of H-bridges (DRV8837, TI) and their corresponding thermoelectric pellets. The MCU can control the corresponding thermoelectric pellets for heating or cooling by controlling the on/off and voltage direction of the H-bridges. The speed of heating or cooling and the final temperature can be controlled by controlling the duty cycle of PWM.

Characterization. The current and voltage data was measured by Data acquisition/multimeter system (DAQ6510, Keithley) with a constant-frequency sampling of 1000 Hz. The temperature change was measured by k-type thermocouples (Kaipusen) and IR camera (FLIR A325sc). SEM images were taken by Field Emission Scanning Electron Microscopy (FESEM, LEO-1530, Zeiss, Germany). The photographs of vibrating mechanical actuators were taken by a high-speed camera (Photron FASTCAM SA4) at a typical recording speed of 2000 fps. The thermal conductivity coefficient of thermal conductive materials was detected by the DRL-III thermal conductivity coefficient meter. The pressure signals were collected by a piezoelectric pressure sensor (FLEXIFORCE). The accelerator signals were collected by a 3-axis accelerometer (ADXL335) in 3-axis.

VR system. We adopted Oculus Rift head-mounted display (HMD), which is a low-cost and widely used VR device. Oculus Rift was equipped with per-eye displays with high-resolution images of 1080 × 1200. Rift has an integrated headset that provides specialized sound effects and has 6 degrees of freedom rotation and position tracking technology. Leap Motion Controller is a small USB peripheral device, which can effectively track the position and motion of the human

hand skeleton in three-dimensional space. At the same time, Leap Motion can also support multiple platforms. It is user-friendly and can help designers develop new applications efficiently. The software is written in C# and the VR applications are implemented in Unity3D Game Engine. Unity 3D is a real-time content development platform that can be used to develop 2D/3D games and applications. It has cross-platform compatibility, various functions, and an abundance of documentation, which is widely used in the VR and AR fields. We used Unity3D to integrate all the application components and displayed the created VR application in real-time on Oculus Rift HMD. To provide users with a more immersive and fascinating experience, we also designed rich interactive functions using Leap Motion Controller. In addition, we use 3d modeling software (such as 3ds Max, Blender, and Zbrush) to design and build the props and environments, then import the built models and animations into the Unity3D.

Data Availability

All data needed to evaluate the conclusions in the papers are present within the manuscript and the Supplementary Information.

Code Availability

Custom code used in the study is available from the corresponding authors upon reasonable request.

Acknowledgments

This work was supported by Research Grants Council of the Hong Kong Special Administrative Region (Grant No. 11215722, 11213721, 11211523), City University of Hong Kong (Grants No. 9667221, 9680322), National Natural Science Foundation of China (Grants No. 62122002, 12072057, 62172346 and 61907037), Shenzhen Science and Technology Innovation Commission (Grant No. SGDX20220530111401011, JCYJ20200109110201713), Innovation and Technology Fund of Innovation and Technology Commission (GHP/095/20GD), in part of InnoHK Project on Project 2.2—AI-based 3D ultrasound imaging algorithm at Hong Kong Centre for Cerebro-Cardiovascular Health Engineering (COCHE), Guangdong Basic and Applied Basic Research Foundation (Project No. 2021A1515011893), the Centre for Applied Computing and Interactive Media (ACIM) of School of Creative Media, City University of Hong Kong. The CityU Contract Research RMGS (Project No. 9239092), and the CityU Donations for Research Projects RMGS (Project No. 9229075), and the Innovation and Technology Fund (Project No. ITS/326/21FP). LiaoNing Revitalization Talents Program (Grant No. XLYC2007196), and Dalian Outstanding Young Talents in Science and Technology (2021RJ06). E.S. acknowledged the support from Shanghai Municipal Science and Technology Major Project (No. 2018SHZDZX01), ZJ Lab, and Shanghai Center for Brain Science and Brain-Inspired Technology.

Author Contributions Statement

X.Y. and Y.H. initiated the concept and proposed the project. Y.H. designed the whole system and collected the overall data. Y.H., J.Z., X.Y., Z.X. and Mr. X.G optimized the structure of feedback device. J.Z and Y.H. designed and characterized control circuits. Y.H., C.Y., D.L., K.Y., Y.G., Z.C. and J.L. fabricated the feedback devices. P.K. and Y.H. set up the VR scenes. Y.H., J.Z., P.K., K.Z. and S.C. designed and organized user studies. Y.H., X.Y., S.C., P.K., K.Z. and K.Y. analyzed the data obtained from user studies. Mr. X.G. and Z.X. led the mechanical, thermal, and electrical modeling and related finite element analysis. Prof. X.G. gave comments on mechanical modelling.

547 Y.H., J.Z., P.K., Mr. X.G., Z.X., K.Z. and X.Y. wrote the manuscript. All other authors contributed
548 to discussing the data and commenting on the final manuscript.

549 **Competing Interests Statement**

550 The authors declare that there is no conflict of interest regarding the publication of this article.

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Figure Legends/Captions

Fig. 1 | Multimode haptic feedback mechanism. **a**, Schematic of neural tactile sensing system and corresponding three stimulation modes. An exploded-view illustration of a device with 16 independently controlled multimodal haptic feedback units, each unit has a mechanical actuator (MA), two pairs of thermoelectric (TE) pellets and one pair of electrotactile (ES) electrodes. **b**, Selectively stimulation on different receptors by three different feedback modes. **c**, Illustration of different tactile sensations and related parameters for feedback. **d**, The force applied by the mechanical actuators and the free vibration amplitude activated at different frequencies. Center line, median; bar height, mean; error bars, SD; $n = 10$ independent tests. **e**, Frequency discrimination results of electrotactile and mechanical vibration from user study. Points, mean values; shadings, SD; $n = 6$ independent tests. **f**, The description of impulse, vibration and dynamic pressure reproduced by electrotactile.

Fig. 2 | Operation of the multimodal haptic patch. **a**, Illustration of the idea that users can feel real-object sensations by using our haptic feedback interface in the VR. Photographs of different arrays haptic feedback units arrays, including 2×4 patch (**b**), 4×4 patch (**c**) and palm form patch (**d**). **e**, 3D graph of the wireless control circuits. **f**, Time-dependent temperature changes under different power density. Solid lines, experimental data; dotted lines, simulation data. **g**, Thermal simulation of cooling effect when devices were attached on the skin with different outer heat dissipation surfaces. **h**, The value of temperature difference at the points of 5 s and 10 s after activation. Data are presented as mean values $\pm$ SD; $n = 5$ independent tests. **i**, User study results of judging the temperature change on the surface with alike real materials ($k = 0 \sim 0.1$ W/mK, $0.1 \sim 1$ W/mK, and >1 W/mK). Different temperature change can be controlled by the power applied on each unit, where the “negative power” represents the cooling mode. **j**, The time-dependent stress curve felt by the 1cm/s moving finger on the surface with roughness of $269 \mu\text{m}$. **k**, The frequency change versus finger moving speed on surface with different roughness. **l**, The stress collected from dynamic simulation when skin slide over different surface with different morphology. **m**, Summary of the roughness sensation of sandpapers in user study with different virtual rough surfaces created in electrotactile mode. Square, mean; center line, median; box limits, upper and lower quartiles; whiskers, $1.5 \times$ interquartile range; points, outliers; $n = 13$ independent spots. **n**, Photographs of four textures used in user studies for distinguishing the patterns created on actuator arrays, and (**o**) the accuracy in discriminating the four textures. **p**, The rating levels of pressure sensation created by electrotactile at different frequency. Pointes, mean values; shadings, SD; $n = 10$ independent tests. **q**, The box plot of the electrotactile frequency that activates dynamic pressure in user study. Square, mean; center line, median; box limits, upper and lower quartiles; whiskers, $1.5 \times$ interquartile range; points, outliers; $n = 10$ independent spots.

Fig. 3 | User study of multidimensional scaling textures. **a**, 9 different materials tested in user study, including fabric, sandpapers, SS, and plastics with different roughness and pattern. The right radar diagrams are the average rating values of real and virtual textures in five dimensions, soft-rough, flat-lumpy, soft-hard, slippery-sticky, and cold-warm obtained from MDS user studies. The box plots of five-dimensional ratings of these 9 textures. Blue shadings, true textures; red shadings, virtual textures. **b**, Smooth-Rough, **c**, Flat-Lumpy, **d**, Soft-Hard, **e**, Slippery-Sticky, **f**, Cold-Warm, where scores for real and virtual textures are listed in pair. Square, mean; center line, median; box limits, upper and lower quartiles; whiskers, $1.5 \times$ interquartile range; points, outliers; $n = 45$

independent tests. Two-way repeated-measures analysis of variance (ANOVA) was performed on data for each modality. The results of user study in distinguishing different materials with only thermal feedback (g), only mechanical feedback (electrotactile and mechanical deformation) (h), and both thermal feedback and mechanical feedback (electrotactile and mechanical deformation) (i).

Fig. 4 | Demonstrations of the multimodal haptic feedback VR systems. **a**, A schematic picture of a girl wears a VR headset for getting visual and auditory feedback. The Leap motion tracks the movement of her hand and send messages to the computer where the control center can transmit commands to the haptic feedback patches for stimulating the user getting haptic feedback on hand and arms when touching different virtual objects in virtual world. **b**, The girl uses an alcohol wet wipe to disinfect the hand, where the temperature decreases due to the evaporation of alcohol and the palm can feel slight force produced by mechanical actuators. Heat maps present the distribution of the temperature change and mechanical vibration amplitude in the palm-form device, where P1~P16 are the positions of 16 multimode haptic units. **c**, The girl touches the surface of a book where the temperature decreases due to the lower book temperature and user's finger feels the roughness activated by electrotactile. Heat maps present the distribution of the temperature change and the electrotactile intensity in the palm-form device. Since the four fingers moved simultaneously, the frequency of stimulation is the same on the four fingertips (P2~P5), corresponding with the same electrical waveforms. **d**, The girl feels the shaking of cloth under strong wind and feels chill due to the heat transfer with haptic feedback patch. The sleeve flapped the arm at the same time, produced by the mechanical actuators. Heat maps present the distribution of the temperature change and mechanical vibration amplitude in the device attached on the arm, where A1~A16 are the positions of 16 multimode haptic units in the 4×4 patch. **e**, The girl feels a worm fell on the arm and crawl. The first impact felt by the arm at the point worm fall on the arm (A5) were activated by the electrotactile, then the temperature decreases slightly due to the heat transfer with worm, and the crawling behavior of the worm is activated by the mechanical actuators (A5, A6 → A6, A7 → A7, A8).

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Fig. 1 | Multimode haptic feedback mechanism.

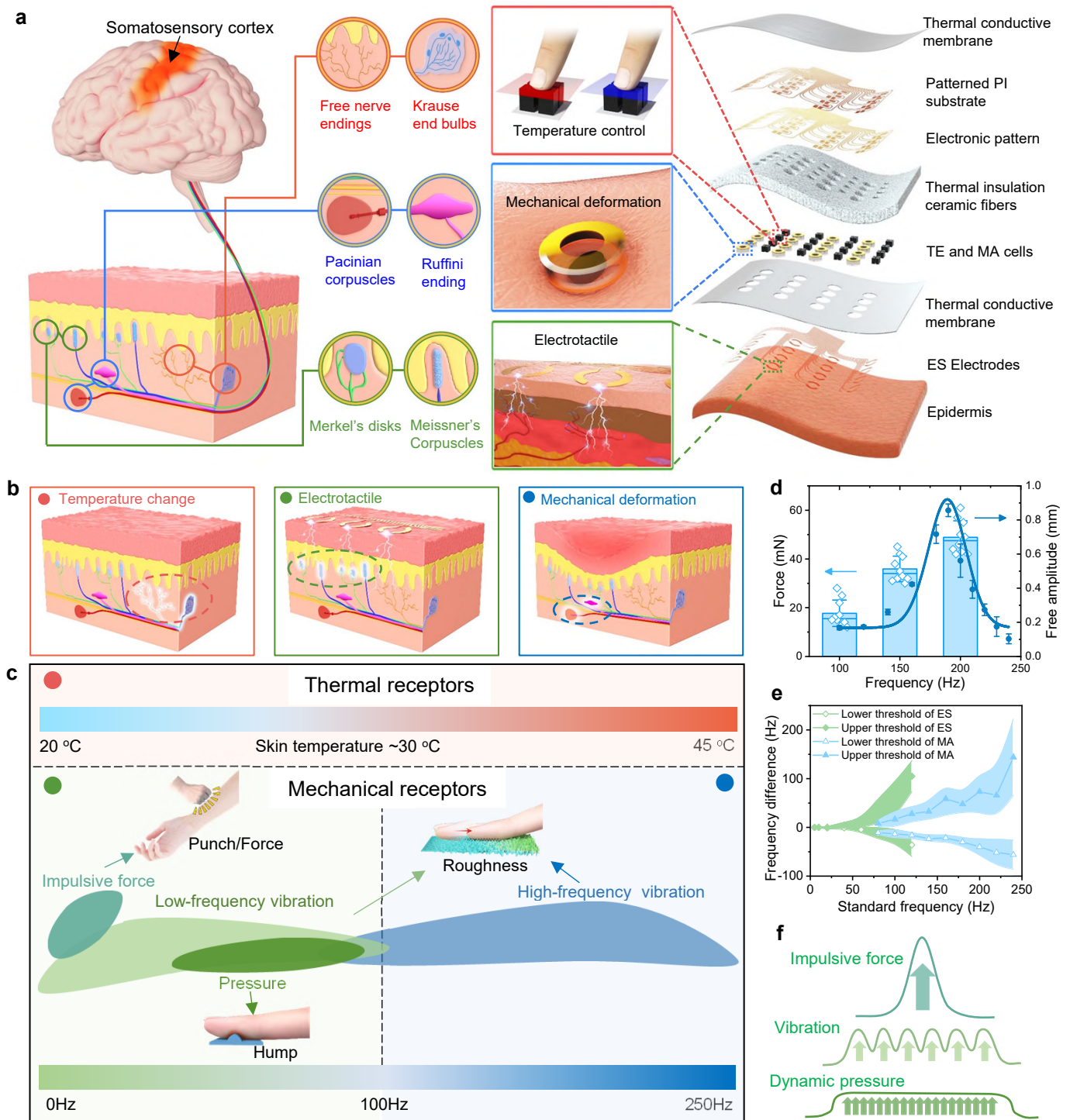


Fig. 2 | Operation of the multimodal haptic patch.

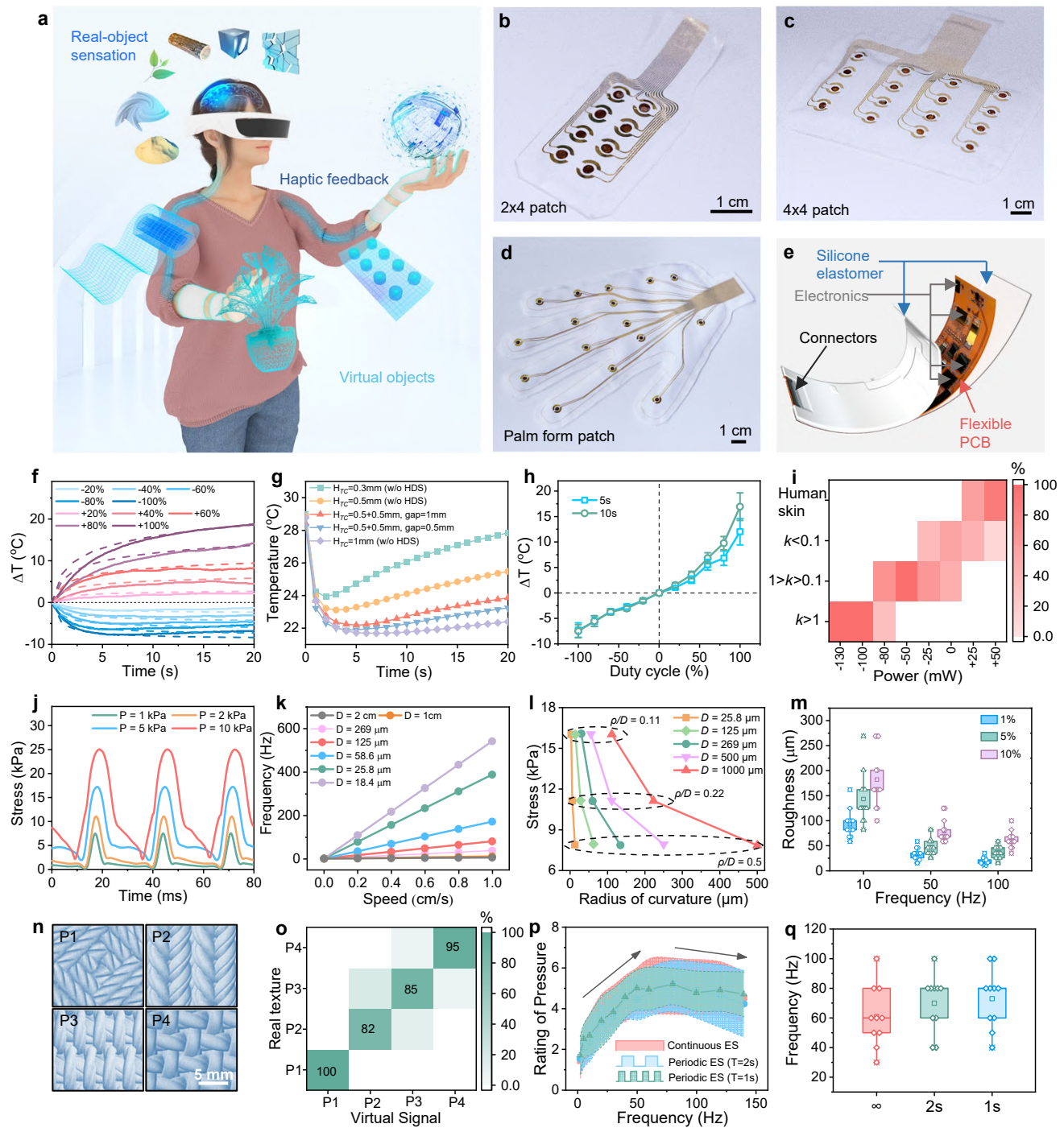


Fig. 3 | User study of multidimensional scaling textures.

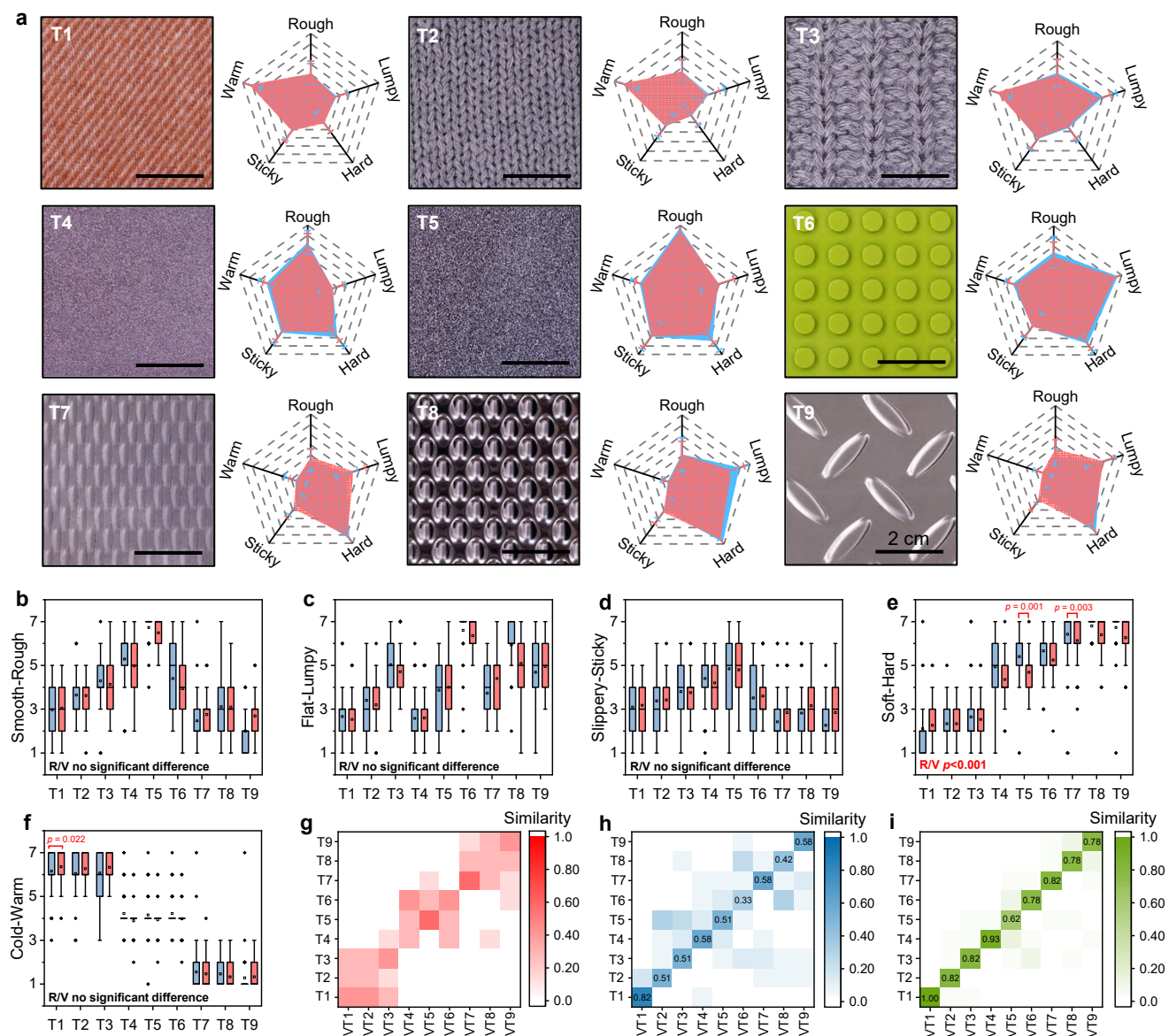


Fig. 4 | Demonstrations of the multimodal haptic feedback VR systems.

